

Market Regime and Risk Lab

Forecasting Five-Day Future Volatility of BTC-USD

Portfolio Research Project

August 2026

Abstract

This report studies whether information available at the end of a day can improve the prediction of the realised volatility of the next five daily BTC-USD observations. The project uses daily OHLCV data from Yahoo Finance, engineered return, volatility and drawdown features, a direct historical-volatility baseline, Ridge regression and Random Forest. The evaluation is chronological and uses expanding-window walk-forward folds. The target boundary is explicitly protected: a training observation is included only when its five-day target ends inside the training period, and a test observation is included only when its target ends inside the evaluated year. Ridge regression achieves lower RMSE than the historical-volatility baseline in all four evaluation years. Random Forest improves on the baseline only in 2022. Nevertheless, all Ridge R^2 values are negative, so the result is a limited and period-dependent improvement rather than evidence of highly accurate forecasting or trading profitability.

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1 Research question

The research question is:

Can information available at the end of date t improve the prediction of BTC-USD realised volatility over the next five daily observations, compared with a simple historical-volatility baseline?

The project is deliberately a forecasting and evaluation study. It does not claim to predict the market perfectly, and it does not include an autonomous trading strategy.

2 Data and data limitations

The dataset contains daily BTC-USD OHLCV data retrieved from Yahoo Finance through the `yfinance` library [1]. The requested period is 1 January 2018 to 31 December 2025; the final available observation is 30 December 2025.

Table 1: Dataset summary

Asset	BTC-USD
Source	Yahoo Finance through <code>yfinance</code>
Frequency	Daily
First observation	2018-01-01
Last observation	2025-12-30
Number of observations	2,921
Raw file	<code>data/raw/btc_data.csv</code>
Processed file	<code>data/processed/btc_features.csv</code>

The CSV export contains three structural metadata rows before the numerical observations. These rows identify the columns, ticker and date level; they are not market observations and are removed during processing. No numerical values are imputed. Dates are sorted chronologically and duplicate dates are checked.

Bitcoin trades continuously, including weekends. Consequently, the horizon is expressed as five daily observations rather than five conventional stock-market trading days. The dataset contains daily aggregates and does not describe the intraday price path. The source timezone is not recorded in the downloaded CSV, and Yahoo Finance volume should be interpreted as provider-reported aggregate volume rather than complete exchange-wide volume.

3 Feature construction

Features are calculated using information available at or before the prediction date t . The processed table contains:

- one-day, five-day and twenty-day returns;
- five-day, twenty-day and sixty-day rolling volatility;
- downside volatility;
- twenty-day and sixty-day moving averages;
- distance from a moving average;
- historical and recent drawdown;

- recent high-low range;
- volume change.

Rolling quantities use past observations only. The rolling-window calculations follow the standard rolling-dataframe approach documented by pandas [4]. Warm-up rows without sufficient history are excluded from modelling. Volatility is calculated from returns rather than raw prices.

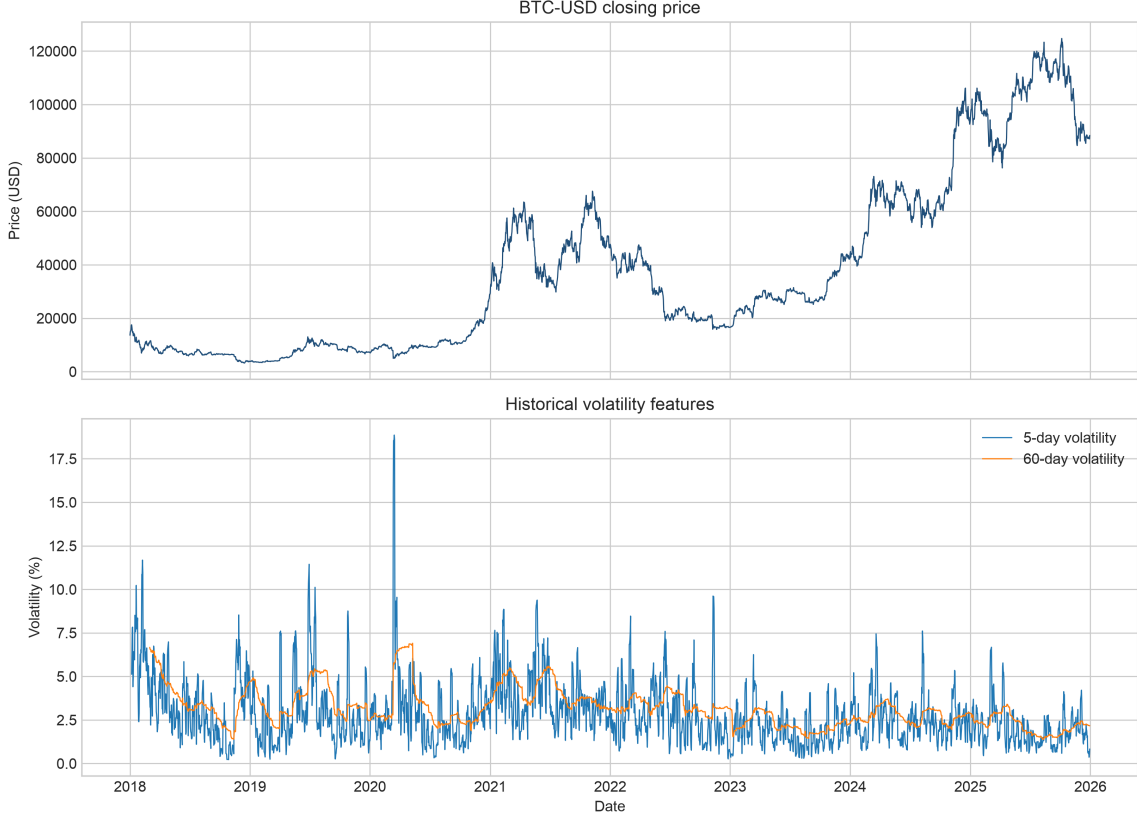


Figure 1: BTC-USD closing price and selected historical-volatility features.

Figure 1 illustrates why volatility is treated as a changing market condition rather than a constant property of the asset. The series includes quiet and highly turbulent periods, which motivates chronological evaluation and robustness checks.

4 Prediction target and leakage control

For each prediction date t , the target is the realised volatility of the next five daily returns:

$$y_t = \text{std}(r_{t+1}, r_{t+2}, r_{t+3}, r_{t+4}, r_{t+5}), \quad (1)$$

where r_i is the daily percentage return. The target is created by shifting the five-day volatility series five observations into the future.

The target has both a prediction date and a target-end date. If a training observation near the end of 2021 has a target window extending into January 2022, it cannot be used in a training fold ending in 2021. The final evaluation therefore stores the target-end date and applies the following rule:

A row is included in a fold only if its complete target window remains inside the period assigned to that fold.

This boundary rule prevents future evaluation-period returns from entering the training targets. The final observations without a complete five-day target are removed.

5 Models and evaluation design

The project compares three approaches.

5.1 Historical-volatility baseline

The baseline uses recent five-day historical volatility as the forecast for future five-day volatility. It is intentionally simple: it tests whether machine-learning models add information beyond a direct recent-volatility estimate.

5.2 Ridge regression

Ridge regression is the main regularised linear model. Regularisation is useful here because return and volatility features can be correlated. The model is fitted with $\alpha = 1$. The project uses explicit numerical feature selection so that dates, raw identifiers and the target cannot accidentally enter the feature matrix. The general scikit-learn framework for composing estimators is documented in [2].

5.3 Random Forest

Random Forest provides a nonlinear tree-based comparison. It tests whether flexible nonlinear relationships improve the forecasts. It is not treated as automatically superior merely because it is more complex.

5.4 Walk-forward evaluation

The final evaluation uses expanding-window chronological folds:

- train through 2021 and evaluate 2022;
- train through 2022 and evaluate 2023;
- train through 2023 and evaluate 2024;
- train through 2024 and evaluate 2025.

Random shuffling is not used for the final evaluation. Time-ordered validation is consistent with the purpose of time-series cross-validation documentation such as scikit-learn's `TimeSeriesSplit` reference [3]. The project also keeps the target-end boundary explicit rather than relying only on the prediction-date boundary.

The main metrics are mean absolute error (MAE), root mean squared error (RMSE), R^2 , Pearson correlation and Spearman rank correlation. MAE is easy to interpret, while RMSE penalises large errors more heavily. Pearson and Spearman correlations describe association, not necessarily good calibration.

6 Results

The values below come from `reports/final_walk_forward_results.csv`, generated by the boundary-safe final evaluation script.

Table 2: Boundary-safe walk-forward results

Year	Model	MAE	RMSE	R^2	Better than baseline
2022	Historical baseline	1.6937	2.2725	-0.7505	Reference
2022	Ridge	1.3308	1.7508	-0.0390	Yes
2022	Random Forest	1.4432	1.8945	-0.2166	Yes
2023	Historical baseline	1.1338	1.4307	-0.7436	Reference
2023	Ridge	1.0302	1.2166	-0.2609	Yes
2023	Random Forest	1.4712	1.8010	-1.7631	No
2024	Historical baseline	1.1579	1.5415	-0.5847	Reference
2024	Ridge	1.0571	1.2703	-0.0761	Yes
2024	Random Forest	1.4290	1.6361	-0.7851	No
2025	Historical baseline	1.0140	1.3309	-0.5390	Reference
2025	Ridge	0.9465	1.1420	-0.1332	Yes
2025	Random Forest	1.2428	1.5309	-1.0362	No

Ridge has the lowest RMSE among the three models in every evaluation year. Its advantage over the baseline is visible in both RMSE and MAE. Random Forest improves on the baseline in 2022 but is worse in the following three years.

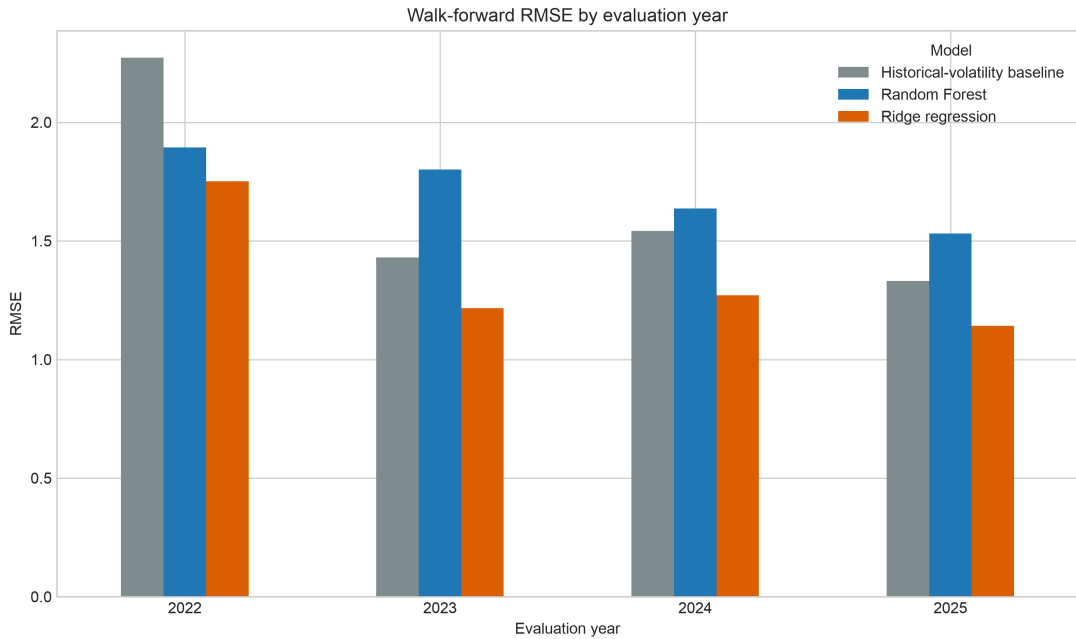


Figure 2: RMSE comparison across the four walk-forward evaluation years.

The negative Ridge R^2 values require careful interpretation. They indicate that Ridge does not beat a constant-average benchmark in the R^2 comparison, even though it beats the historical-volatility baseline in this experiment. Therefore, the result is a relative improvement over one practical baseline, not strong absolute predictive accuracy.

The Pearson and Spearman correlations are positive but modest. The predictions contain some association with future volatility, especially in 2025, but the relationships are not strong

enough to support claims of reliable forecasting in all regimes.



Figure 3: Actual and predicted five-day future volatility by evaluation year.

7 Robustness checks

7.1 Alternative volatility windows

Ridge was tested with five-day, twenty-day and sixty-day volatility specifications. The sixty-day specification produced the lowest yearly MSE among the tested variants in the earlier sensitivity experiments, but the differences were modest. This supports reporting the longer window as a useful sensitivity result rather than claiming it is universally optimal.

7.2 Reduced feature set

A reduced Ridge specification used only one-day return, five-day return, five-day volatility, twenty-day volatility, sixty-day volatility and drawdown. It still beat the historical-volatility baseline in each yearly walk-forward period, but it was not consistently better than the full Ridge feature set. This suggests that the main relative advantage of Ridge is not entirely dependent on every engineered feature, while the exact performance remains feature- and period-dependent.

7.3 Excluding 2020

The 2020 sensitivity is now applied to every walk-forward training fold. The purpose is to test whether the extreme 2020 volatility regime controls the conclusion. Removing 2020 slightly changes the Ridge results: it worsens performance in 2022–2024 and slightly improves the 2025 fold. The effect is therefore period-dependent rather than uniformly beneficial or harmful.

8 Decision simulation

A decision simulation is not included in this version. The forecasting experiment should be understood separately from a trading strategy. A future extension could study a simple exposure overlay, but it would need an explicit execution delay, position limit, cash treatment, transaction cost, slippage, turnover, leverage assumption and drawdown analysis. Forecast quality would still have to be reported separately from strategy performance.

9 Limitations

The main limitations are:

1. the study uses one asset, so the results may not generalise to other crypto-assets or financial markets;
2. daily OHLCV data does not represent the intraday path;
3. the source timezone is not recorded in the CSV;
4. Yahoo Finance volume is provider-reported aggregate volume;
5. Bitcoin's market structure and volatility are non-stationary;
6. the sample contains only four yearly evaluation folds;
7. all Ridge R^2 values are negative;
8. no transaction costs or execution effects are included because no strategy is simulated;
9. the final data may be revised if Yahoo Finance is queried again.

The evidence is therefore insufficient to claim general profitability or a production-ready forecasting system. The contribution of the project is methodological: it demonstrates a reproducible, leakage-aware and honest financial machine-learning experiment.

10 Reproducibility

The project is organised as follows:

- **data/raw/**: unchanged downloaded data;
- **data/processed/**: engineered features;
- **scripts/**: data collection, feature engineering, exploration and modelling scripts;
- **reports/**: result tables, dated predictions, figures and bibliography;
- **docs/**: the detailed project guide.

From the repository root, the main workflow is:

```
.venv/bin/python 'projects/Market Regime and Risk Lab/scripts/download_data.py'  
.venv/bin/python 'projects/Market Regime and Risk Lab/scripts/create_features.py'  
.venv/bin/python 'projects/Market Regime and Risk Lab/scripts/final_evaluation.py'
```

The report figures are generated from the processed dataset and the final prediction CSV. The final evaluation writes the comparison table to **reports/final_walk_forward_results.csv** and dated predictions to **reports/final_walk_forward_predictions.csv**.

11 Conclusion

This project finds that Ridge regression improves on a simple historical-volatility baseline in all four boundary-safe yearly walk-forward periods. Random Forest is less stable and performs worse than the baseline from 2023 onward. The feature-window and feature-reduction checks show that the Ridge comparison is reasonably stable, but its absolute predictive power remains limited. The negative R^2 values are especially important: the model should not be described as a highly accurate volatility predictor.

The most defensible conclusion is:

Under the specific BTC-USD sample, feature design and chronological evaluation used here, Ridge regression provides a limited and period-dependent improvement over the historical-volatility baseline. The evidence does not establish universal forecasting accuracy or trading profitability.

References

- [1] Ran Aroussi. `yfinance`: Yahoo finance market data downloader. GitHub repository, 2025. Accessed 2026-08-07.
- [2] scikit-learn developers. Pipelines and composite estimators, 2025. Accessed 2026-08-07.
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- [4] The pandas development team. `pandas.dataframe.rolling` documentation, 2025. Accessed 2026-08-07.